(https://databricks.com)

Ingest races file

```
dbutils.widgets.text("p_data_source", "")
v_data_source = dbutils.widgets.get("p_data_source")
```

```
dbutils.widgets.text("p_file_date", "2021-03-21")
v_file_date = dbutils.widgets.get("p_file_date")
```

```
%run "../includes/configuration"
```

```
%run "../includes/common_functions"
```

```
from pyspark.sql.types import StructType, StructField, IntegerType, StringType, DateType
```

Step1 - Creating schema and reading data intop the schema

```
races_schema = StructType(fields=[StructField("raceId", IntegerType(), False),
                                  StructField("year", IntegerType(), True),
                                  StructField("round", IntegerType(), True),
                                  StructField("circuitId", IntegerType(), True),
                                  StructField("name", StringType(), True),
                                  StructField("date", DateType(), True),
                                  StructField("time", StringType(), True),
                                  StructField("url", StringType(), True)
                                  ])
```

```
races_df = spark.read \
    .option("header", True) \
    .schema(races_schema) \
    .csv(f"{raw_folder_path}/{v_file_date}/races.csv")
```

```
display(races_df)
```

Table

	raceId	year	round	circuitId	name	date	time	url
1	1	2009	1	1	Australian Grand Prix	2009-03-29	06:00:00	http://en.wikipedia.org
2	2	2009	2	2	Malaysian Grand Prix	2009-04-05	09:00:00	http://en.wikipedia.org
3	3	2009	3	17	Chinese Grand Prix	2009-04-19	07:00:00	http://en.wikipedia.org

4	4	2009	4	3	Bahrain Grand Prix	2009-04-26	12:00:00	http://en.wikipedia.org
5	5	2009	5	4	Spanish Grand Prix	2009-05-10	12:00:00	http://en.wikipedia.org
6	6	2009	6	6	Monaco Grand Prix	2009-05-24	12:00:00	http://en.wikipedia.org
7	7	2009	7	5	Turkish Grand Prix	2009-06-07	12:00:00	http://en.wikipedia.org
8	8	2009	8	9	British Grand Prix	2009-06-21	12:00:00	http://en.wikipedia.org
9	9	2009	9	20	German Grand Prix	2009-07-12	12:00:00	http://en.wikipedia.org
10	10	2009	10	11	Hungarian Grand Prix	2009-07-26	12:00:00	http://en.wikipedia.org
11	11	2009	11	12	European Grand Prix	2009-08-23	12:00:00	http://en.wikipedia.org
12	12	2009	12	13	Belgian Grand Prix	2009-08-30	12:00:00	http://en.wikipedia.org
13	13	2009	13	14	Italian Grand Prix	2009-09-13	12:00:00	http://en.wikipedia.org
14	14	2009	14	15	Singapore Grand Prix	2009-09-27	12:00:00	http://en.wikipedia.org

1,058 rows

Step 2 - Add ingestion_date and race_timestamp columns to the dataframe

```
from pyspark.sql.functions import current_timestamp, to_timestamp, lit, concat, col
```

```
races_with_timestamp_df = races_df.withColumn("ingestion_date", current_timestamp()) \
                          .withColumn("race_timestamp", to_timestamp(concat(col('date'), lit(' '), col('time')), 'yyyy-MM-dd HH:m
                          .withColumn("data_source", lit(v_data_source)) \
                          .withColumn("file_date", lit(v_file_date))
```

```
display(races_with_timestamp_df)
```

Table

	raceId	year	round	circuitId	name	date	time	url
1	1	2009	1	1	Australian Grand Prix	2009-03-29	06:00:00	http://en.wikipedia.org
2	2	2009	2	2	Malaysian Grand Prix	2009-04-05	09:00:00	http://en.wikipedia.org
3	3	2009	3	17	Chinese Grand Prix	2009-04-19	07:00:00	http://en.wikipedia.org
4	4	2009	4	3	Bahrain Grand Prix	2009-04-26	12:00:00	http://en.wikipedia.org
5	5	2009	5	4	Spanish Grand Prix	2009-05-10	12:00:00	http://en.wikipedia.org
6	6	2009	6	6	Monaco Grand Prix	2009-05-24	12:00:00	http://en.wikipedia.org
7	7	2009	7	5	Turkish Grand Prix	2009-06-07	12:00:00	http://en.wikipedia.org
8	8	2009	8	9	British Grand Prix	2009-06-21	12:00:00	http://en.wikipedia.org
9	9	2009	9	20	German Grand Prix	2009-07-12	12:00:00	http://en.wikipedia.org
10	10	2009	10	11	Hungarian Grand Prix	2009-07-26	12:00:00	http://en.wikipedia.org
11	11	2009	11	12	European Grand Prix	2009-08-23	12:00:00	http://en.wikipedia.org
12	12	2009	12	13	Belgian Grand Prix	2009-08-30	12:00:00	http://en.wikipedia.org
13	13	2009	13	14	Italian Grand Prix	2009-09-13	12:00:00	http://en.wikipedia.org
14	14	2009	14	15	Singapore Grand Prix	2009-09-27	12:00:00	http://en.wikipedia.org

1,058 rows

```
races_with_ingestion_date_df = add_ingestion_date(races_with_timestamp_df)
```

Step 3 - Select only the required columns and renaming columns as required

```
races_selected_df = races_with_timestamp_df.select(col('raceId').alias('race_id'), col('year').alias('race_year'),col('round'),
                                                   col('circuitId').alias('circuit_id'),col('name'),col('ingestion_date'),col('race
```

```
display(races_selected_df)
```

Table

	race_id	race_year	round	circuit_id	name	ingestion_date	race_timest
1	1	2009	1	1	Australian Grand Prix	2024-07-24T14:29:58.534+00:...	2009-03-29T06:
2	2	2009	2	2	Malaysian Grand Prix	2024-07-24T14:29:58.534+00:...	2009-04-05T09:
3	3	2009	3	17	Chinese Grand Prix	2024-07-24T14:29:58.534+00:...	2009-04-19T07:
4	4	2009	4	3	Bahrain Grand Prix	2024-07-24T14:29:58.534+00:...	2009-04-26T12:
5	5	2009	5	4	Spanish Grand Prix	2024-07-24T14:29:58.534+00:...	2009-05-10T12:
6	6	2009	6	6	Monaco Grand Prix	2024-07-24T14:29:58.534+00:...	2009-05-24T12:
7	7	2009	7	5	Turkish Grand Prix	2024-07-24T14:29:58.534+00:...	2009-06-07T12:
8	8	2009	8	9	British Grand Prix	2024-07-24T14:29:58.534+00:...	2009-06-21T12:
9	9	2009	9	20	German Grand Prix	2024-07-24T14:29:58.534+00:...	2009-07-12T12:
10	10	2009	10	11	Hungarian Grand Prix	2024-07-24T14:29:58.534+00:...	2009-07-26T12:
11	11	2009	11	12	European Grand Prix	2024-07-24T14:29:58.534+00:...	2009-08-23T12:
12	12	2009	12	13	Belgian Grand Prix	2024-07-24T14:29:58.534+00:...	2009-08-30T12:
13	13	2009	13	14	Italian Grand Prix	2024-07-24T14:29:58.534+00:...	2009-09-13T12:
14	14	2009	14	15	Singapore Grand Prix	2024-07-24T14:29:58.534+00:...	2009-09-27T12:

1,058 rows

Step 4 - Write output to processed container in parquet format

```
# races_selected_df.write.mode("overwrite").format("parquet").saveAsTable("f1_processed.races")
```

```
races_selected_df.write.mode("overwrite").partitionBy('race_year').format("delta").saveAsTable("f1_processed.races")
```

```
%fs
ls /mnt/formula1dl1204/processed/races
```

Table

	path	name	size	modificationTime
1	dbfs:/mnt/formula1dl1204/processed/races/_delta_log/	_delta_log/	0	1721831399000
2	dbfs:/mnt/formula1dl1204/processed/races/race_year=195...	race_year=195...	0	1721831400000
3	dbfs:/mnt/formula1dl1204/processed/races/race_year=195...	race_year=195...	0	1721831400000
4	dbfs:/mnt/formula1dl1204/processed/races/race_year=195...	race_year=195...	0	1721831400000
5	dbfs:/mnt/formula1dl1204/processed/races/race_year=195...	race_year=195...	0	1721831400000
6	dbfs:/mnt/formula1dl1204/processed/races/race_year=195...	race_year=195...	0	1721831400000
7	dbfs:/mnt/formula1dl1204/processed/races/race_year=195...	race_year=195...	0	1721831400000
8	dbfs:/mnt/formula1dl1204/processed/races/race_year=195...	race_year=195...	0	1721831400000

9	dbfs:/mnt/formula1dl1204/processed/races/race_year=195...	race_year=195...	0	1721831400000
10	dbfs:/mnt/formula1dl1204/processed/races/race_year=195...	race_year=195...	0	1721831400000
11	dbfs:/mnt/formula1dl1204/processed/races/race_year=195...	race_year=195...	0	1721831400000
12	dbfs:/mnt/formula1dl1204/processed/races/race_year=196...	race_year=196...	0	1721831401000
13	dbfs:/mnt/formula1dl1204/processed/races/race_year=196...	race_year=196...	0	1721831401000
14	dbfs:/mnt/formula1dl1204/processed/races/race_year=196...	race_year=196...	0	1721831401000
15	dbfs:/mnt/formula1dl1204/processed/races/race_year=196...	race_year=196...	0	1721831401000

73 rows

```
# df = spark.read.parquet(f'{processed_folder_path}/races')
# df
```

```
%sql
select * from f1_processed.races;
```

Table

	race_id	race_year	round	circuit_id	name	ingestion_date	race_time
1	1053	2021	2	21	Emilia Romagna Grand P...	2024-07-24T14:29:59.841+00:...	2021-04-18T
2	1052	2021	1	3	Bahrain Grand Prix	2024-07-24T14:29:59.841+00:...	2021-03-28T
3	1051	2021	21	1	Australian Grand Prix	2024-07-24T14:29:59.841+00:...	2021-11-21T
4	1054	2021	3	20	TBC	2024-07-24T14:29:59.841+00:...	null
5	1055	2021	4	4	Spanish Grand Prix	2024-07-24T14:29:59.841+00:...	2021-05-09T
6	1056	2021	5	6	Monaco Grand Prix	2024-07-24T14:29:59.841+00:...	2021-05-23T
7	1057	2021	6	73	Azerbaijan Grand Prix	2024-07-24T14:29:59.841+00:...	2021-06-06T
8	1058	2021	7	7	Canadian Grand Prix	2024-07-24T14:29:59.841+00:...	2021-06-13T
9	1059	2021	8	34	French Grand Prix	2024-07-24T14:29:59.841+00:...	2021-06-27T
10	1060	2021	9	70	Austrian Grand Prix	2024-07-24T14:29:59.841+00:...	2021-07-04T
11	1061	2021	10	9	British Grand Prix	2024-07-24T14:29:59.841+00:...	2021-07-18T
12	1062	2021	11	11	Hungarian Grand Prix	2024-07-24T14:29:59.841+00:...	2021-08-01T
13	1063	2021	12	13	Belgian Grand Prix	2024-07-24T14:29:59.841+00:...	2021-08-29T
14	1064	2021	13	39	Dutch Grand Prix	2024-07-24T14:29:59.841+00:...	2021-09-05T

1,058 rows

```
dbutils.notebook.exit("Success")
```

```
Success
```

Optional Step - Partitioning data based on race_year

```
#races_selected_df.write.mode("overwrite").partitionBy('race_year').parquet('/mnt/formula1dl1204/processed/races')
```